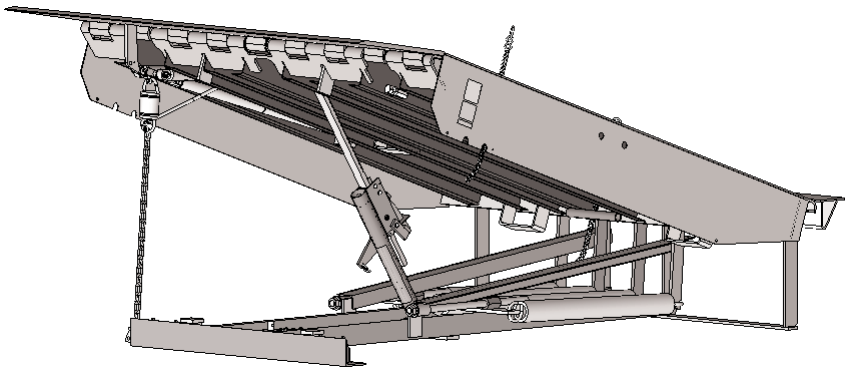
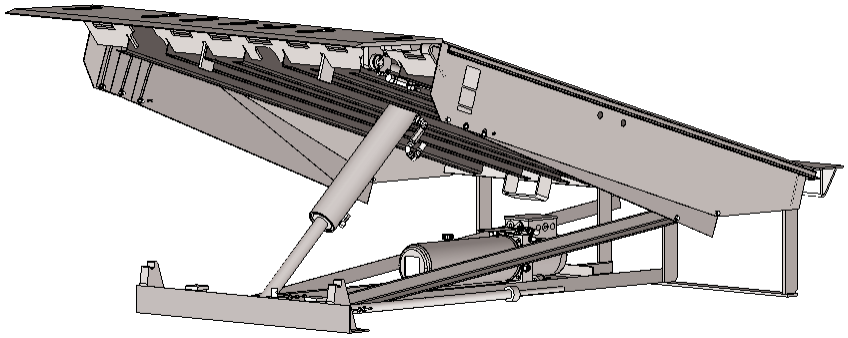
RITE HITE®

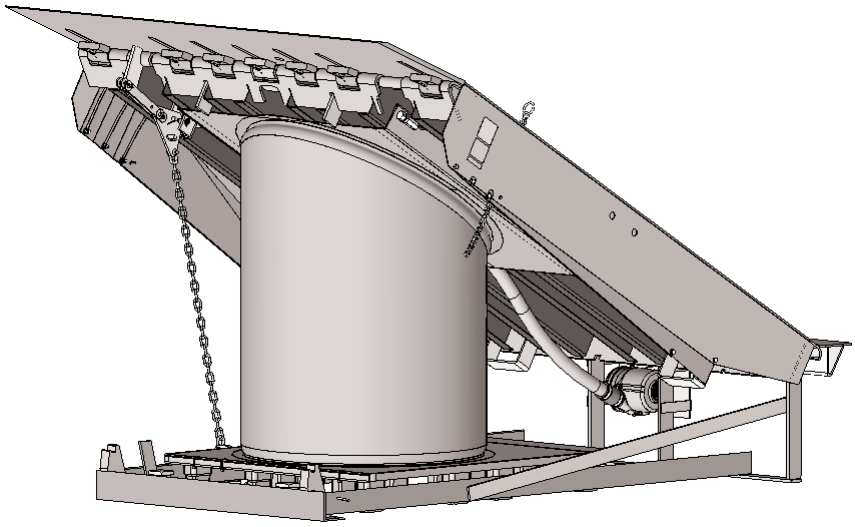
Dock Leveler Installation Manual



RHM Mechanical Dock Leveler

RHH Hydraulic Hydra-Rite™ / Base Dock Leveler





RHA Air Powered Dock Leveler

MADE IN U.S.A.

This Manual Covers Levelers Built After Serial Numbers: **Effective 3/10/06**
RHM Series: 04ID410001M and up **RHH Series:** 05BD480001M and up
RHA Series: 04ID400001M and up

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NOTES

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NOTICE TO USER

Your local RITE-HITE® representative provides a Planned Maintenance Program (P.M.P.) which can be fitted to your specific operation. Call your local representative or the RITE-HITE® Corporation at 414-355-2600.

The RITE-HITE® products in this manual are covered by one or more of the following U.S. patents: 4,744,121; 4,819,770; 4,843,373; 4,865,507; 4,920,598; 4,995,130; 5,040,258; 5,111,546; 5,212,846; 5,271,183; 5,299,386; 5,311,628; 5,323,503; 5,375,965; 5,440,772; 5,442,825; 5,453,735; 5,531,557; 5,546,623; 5,553,987; 5,582,498; 5,664,930; 5,702,223; 5,762,459 (RE: 37,570); 5,882,167; 5,964,572; 6,010,297; 6,052,268; 6,065,172; 6.070,283; 6,074,157; 6,085,375; 6,092,970; 6,106,212; 6,116,839; 6,190,109; 6,220,809; 6,627,016; 6,238,163; 6,311,352; 6,322,310; 6,360,394; 6,368,043; 6,431,819; 6,488.464; 6,497,067; 6,499,169; 6,505,713; 6,524,053; 6,634,049; 6,654,976; 6,676,360; 6,726,432; 6,773,221 and pending U.S and foreign patent applications. RITE-HITE®, LEVEL-RITE®, THINMAN™, SAFE-T-LIP®, HYDRACHEK®, WHEEL-LOK™, DOK-LOK®, DUAL-DOK®, SAFE-T-STRUT™, DOK-COMMANDER®, JUMBO™, HYDRA-RITE™ and SAFE-T-GATE® are trademarks of RITE-HITE® Corporation.

SAFETY WARNING

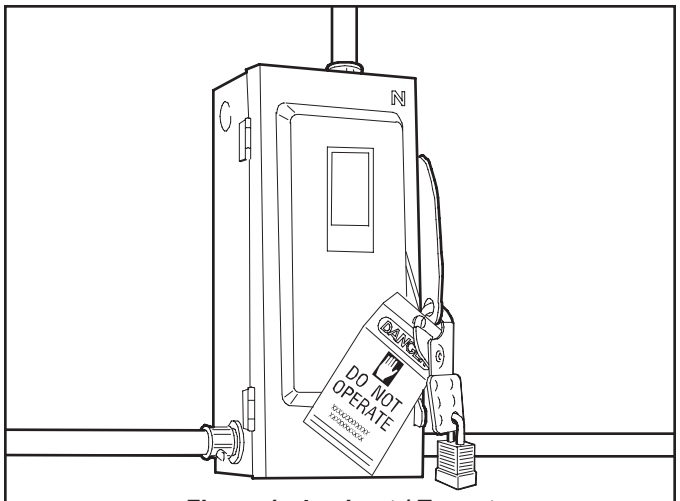


Figure 1 - Lockout / Tagout

⚠ WARNING

When working with electrical or electronic controls, make sure that the power source has been locked out and tagged according to OSHA regulations and approved local electrical codes.

LOCKOUT/TAGOUT PROCEDURES

The Occupational Safety and Health Administration (OSHA) requires that, in addition to posting safety warnings and barricading the work area, the power supply has been locked in the OFF position or disconnected. It is mandatory that an approved lockout device is utilized. An example of a lockout device is illustrated. The proper lockout procedure requires that the person responsible for the repairs is the only person who has the ability to remove the lockout device.

In addition to the lockout device, it is also a requirement to tag the power control in a manner that will clearly note that repairs are under way and state who is responsible for the lockout condition. Tagout devices have to be constructed and printed so that exposure to weather conditions or wet and damp locations will not cause the tag to deteriorate or become unreadable.

RITE-HITE PRODUCTS CORPORATION does not recommend any particular lockout device, but recommends the utilization of an OSHA approved device (refer to OSHA regulation 1910.147). RITE-HITE PRODUCTS CORPORATION also recommends the review and implementation of an entire safety program for the Control of Hazardous Energy (Lockout/Tagout). These regulations are available through OSHA publication 3120.

⚠ DANGER

This is the highest level statement. Failure to follow the listed instructions will most likely result in severe injury or death.

⚠ CAUTION

The statements used with this level of warning deal with a safe operating procedure. If the procedure is ignored, the possibility of personal injury may exist.

⚠ WARNING

This is a statement of serious hazard. Failure to follow the listed instructions could place the individual at risk of serious injury or death.

IMPORTANT

IMPORTANT is used to draw attention to a procedure that needs to be followed to prevent machine or property damage.

OTHER IMPORTANT OPERATIONAL SAFETY WARNINGS

⚠ DANGER

Never be under the dock leveler platform or lip without:

- **Installing the Safe-T-Strut™ or other supporting device.**
- **If lip needs to be extended, follow procedures shown under Safety Devices on the following page.**
- **Turning off power to the control box.**
- **Locking out and tagging out the main power source, as shown under Safety Warnings on preceding page.**

⚠ WARNING

- **Before starting installation or maintenance, check and follow the safety procedures of the facility where the dock leveler is being installed.**
- **Never enter a truck/trailer until its brakes are set, air has been dumped from air ride suspension (if applicable), and you have visually inspected to be sure truck/trailer is securely held in place by a vehicle restraint or wheel chock per OSHA regulations.**
- **Never operate the leveler with you, anyone, or anything on, or in front of the leveler, or without a truck/trailer parked in position, or from on the truck/trailer bed.**
- **DO NOT operate with anyone under platform or in front of the lip.**
- **When leveler is not in use, always store it so that it is supported by the lip supports and that it is level with the surrounding dock floor.**
- **If a malfunction does occur, always call your authorized RITE-HITE® service representative immediately.**

⚠ WARNING

Always barricade the dock leveler at ground level and dock level from any form of traffic when maintenance is required.

⚠ CAUTION

Inspect the dock leveler monthly to ensure that there are no broken or worn parts which could cause injury to personnel or damage to the equipment.

SAFETY DEVICES

⚠ DANGER

Never be under the dock leveler platform or lip without:

- **Installing the Safe-T-Strut™. See below right. This can be done with the assistance of another person by:**
  - **Raise leveler until platform reaches its highest position and lip extends. Continue to maintain this position.**
  - **Then have assistant insert the smaller end of the Safe-T-Strut™ through the hole in the middle of the leveler lip and place the strut's wider open end over the base located on the leveler's front frame. Align the holes on the base and the Safe-T-Strut™ so that the leveler may be secured with the retaining pin and safety clip.**
  - **Release the push button on powered levelers allowing the Safe-T-Strut™ to rest on the underside of the lip.**
- **Lockout/Tagout power supply.**
  - **Turn off the power to the control box.**
  - **Lockout/tagout the main power source, as shown under Safety Warnings on the inside front cover of this manual.**
  - **Always barricade the leveler at dock level and drive level to prevent any unauthorized use of the leveler.**

Remove the Maintenance Support.

- **For Safe-T-Strut™ removal, have an assistant raise the leveler to its highest position with lip fully extended. Release the safety clip and remove retaining pin. Lift strut off base, and remove from lip. Return the Safe-T-Strut™ to the proper storage position.**
- **If you are unable to install the Maintenance Support properly, contact your authorized RITE-HITE® Service Representative or RITE-HITE® Customer Service at 1-414-355-2600.**

⚠ CAUTION

- **Post warnings and barricades at dock level and at drive level to indicate that work is being done around and under the leveler platform.**
- **Lockout/Tagout power to the leveler and post warnings when work is being performed on the leveler.**

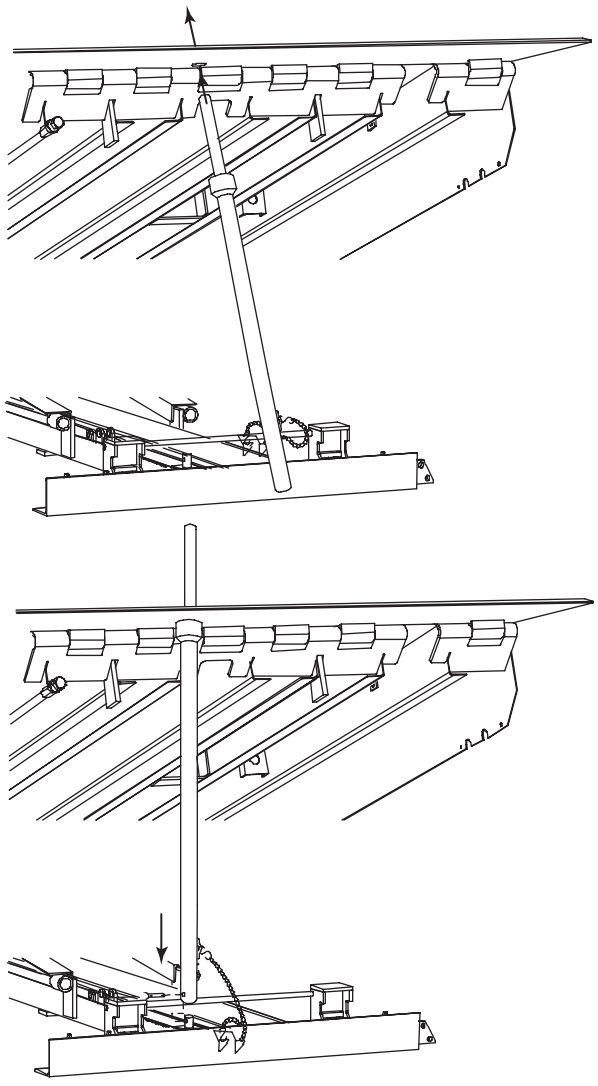
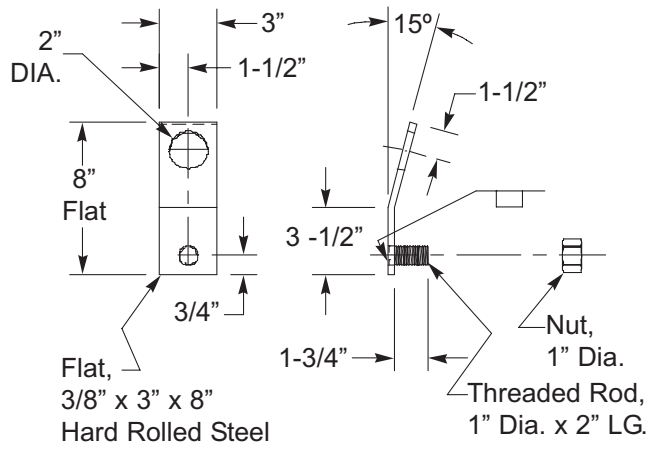


Figure 2 - Install Safe-T-Strut™ Support

INSTALLATION INSTRUCTIONS

PREPARATION PRIOR TO INSTALLATION

1. Check pit to insure it meets pit details. If pit does not match pit details, contact your local representative or RITE-HITE® before proceeding.
   NOTE: Unit may not operate properly if pit is not correct.
2. Leave shipping bands on leveler until instructed to remove them.
3. **RHH and RHA Series Only:** Verify that leveler voltage matches the voltage supplied at pit, and that proper electrical junction box or weatherproof outlet has been installed.
4. Clean out pit and remove all debris from work area.



2 required, Supplied by others.

Figure - 5 Handling Plate

INSTALLATION PROCEDURES

1. Handling plates must be inserted into the forward handling holes of the toe guards. See Figure 5.
2. Attach lifting chain, lift leveler and position above pit. Lower leveler into pit. See Figure 6.
3. Center leveler in pit so that equal gaps are maintained on each side of the leveler. See Figure 7.

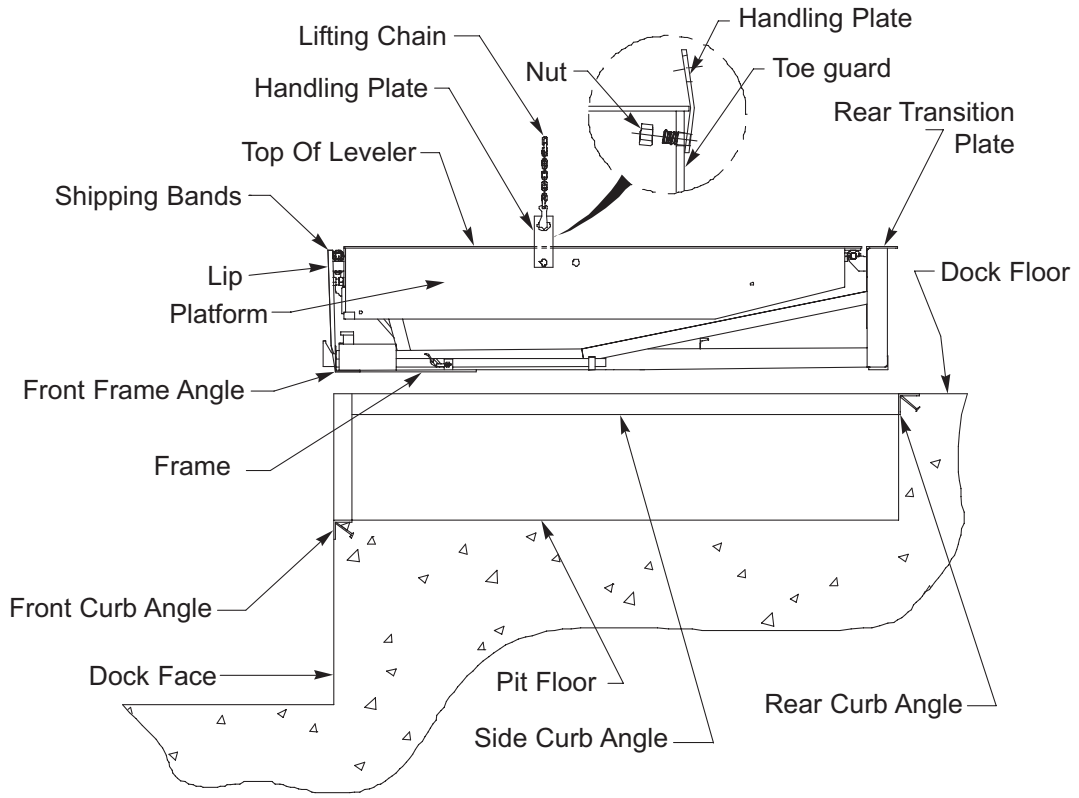


Figure - 6 Placement of Leveler

INSTALLATION INSTRUCTIONS CONT.

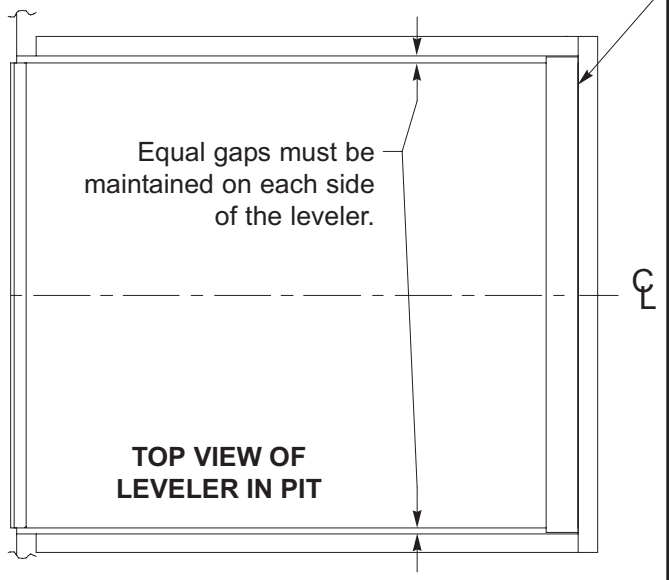


Figure - 7 Leveler Placement

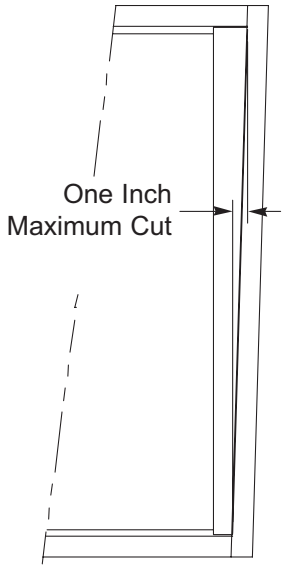


Figure - 8 Trim Transition Plate

4. Push Leveler straight back until the front frame angle is flush with the front curb angle. If the pit is not square, push leveler straight back until the front frame angle is completely behind the front curb angle by no more then 1/4 inch. The front frame angle should never extend past the front curb angle. If the leveler cannot be pushed back far enough, the back edge of the rear transition plate can be cut off up to 1 inch. If trimming the back edge of angle does not allow frame to be positioned properly in pit, contact your local representative or RITE-HITE® for assistance. See Figure 8.
5. Check the area between the rear transition plate and the back of the pit. If a gap exists, place spacers between the rear transition plate and the rear curb angle. Tack weld spacers to the rear transition plate only. See Figure 9.

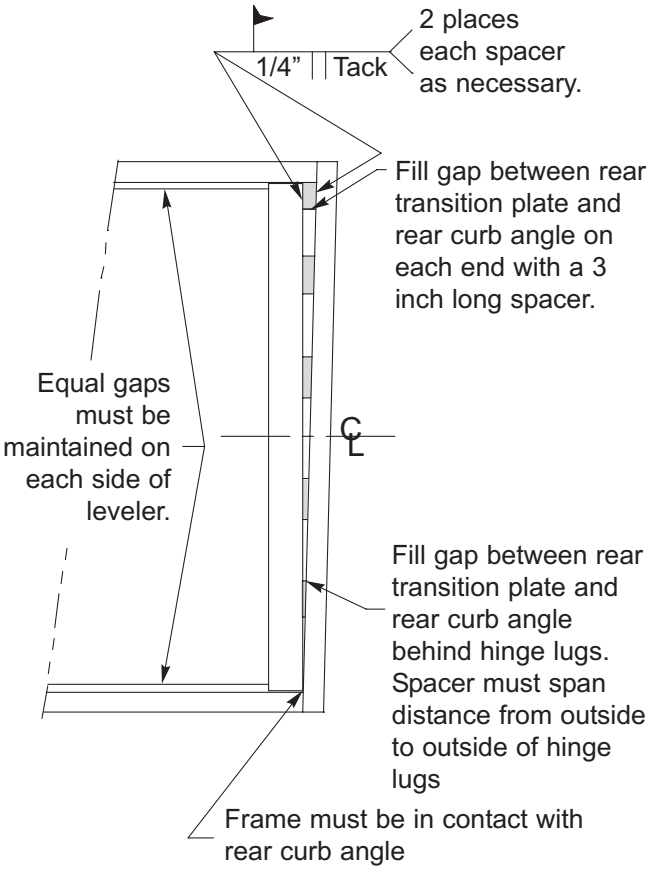


Figure - 9 Rear Frame Spacers

INSTALLATION INSTRUCTIONS CONT.

IMPORTANT

- **ALWAYS ground to the frame or front pit angle when welding. DO NOT ground welding equipment to any electrical components. DO NOT ground to leveler platform when welding on leveler frame. Damage to bearings or electrical components may result.**
- **DO NOT connect green ground lead into control box or junction box until all welding is complete.**
- **Protect Air Tower, Motors, Hose and Hydraulic Lines from damage due to welding, grinding or sharp objects.**

6. Remove lifting chain from handling plates.
7. Lift front frame of leveler until platform is level with dock floor. Cut and place shims under below dock stops, located on front frame angle and maintenance support base. Tack shims together and to front frame angle. See Figure 10.

   IMPORTANT: For specific shimming details see Figure 11.

   NOTE: For levelers with 20 inch lips, lift bottom of front frame angle until it is at 19-1/2 inches below dock floor, and add shims. See Figure 10.
8. Remove and discard all shipping bands.

⚠ CAUTION

Shipping bands are under tension. Wear safety glasses and protective gloves. Stand to one side when cutting bands to avoid being cut by the bands.

9. Place a mark on the face of the front frame angle to indicate the frame centerline. Also, place a mark on the face of the lip to indicate the lip/platform centerline.
10. Shift the front frame angle sideways so that the frame centerline matches up with the lip/platform centerline (using the lines drawn earlier). Once frame and platform have been centered in the pit, tack weld the front frame angle and shims to the front curb angle.
11. **Mechanical leveler:** Pull and hold the release chain to raise the platform. Install maintenance support.

    Powered leveler: Manually raise platform and extend lip on powered levelers. Install maintenance support.
12. Remove handling plates from platform.
13. Lift rear frame of leveler until platform is level with dock floor. Cut and place shims under the rear frame support posts. Tack shims together and to rear frame angle. See Figure 10.

    Mechanical leveler: Be sure to shim underneath the snubbing cable mount on front frame angle. See Figures 10 & 11.

    Hydraulic leveler (Base Only): Place two shims underneath the main cylinder clevis as shown. See Figures 10 & 11.

    IMPORTANT: Insure all shims are in place at this time. Tack shims together and to frame. See Figure 11.

INSTALLATION INSTRUCTIONS CONT.

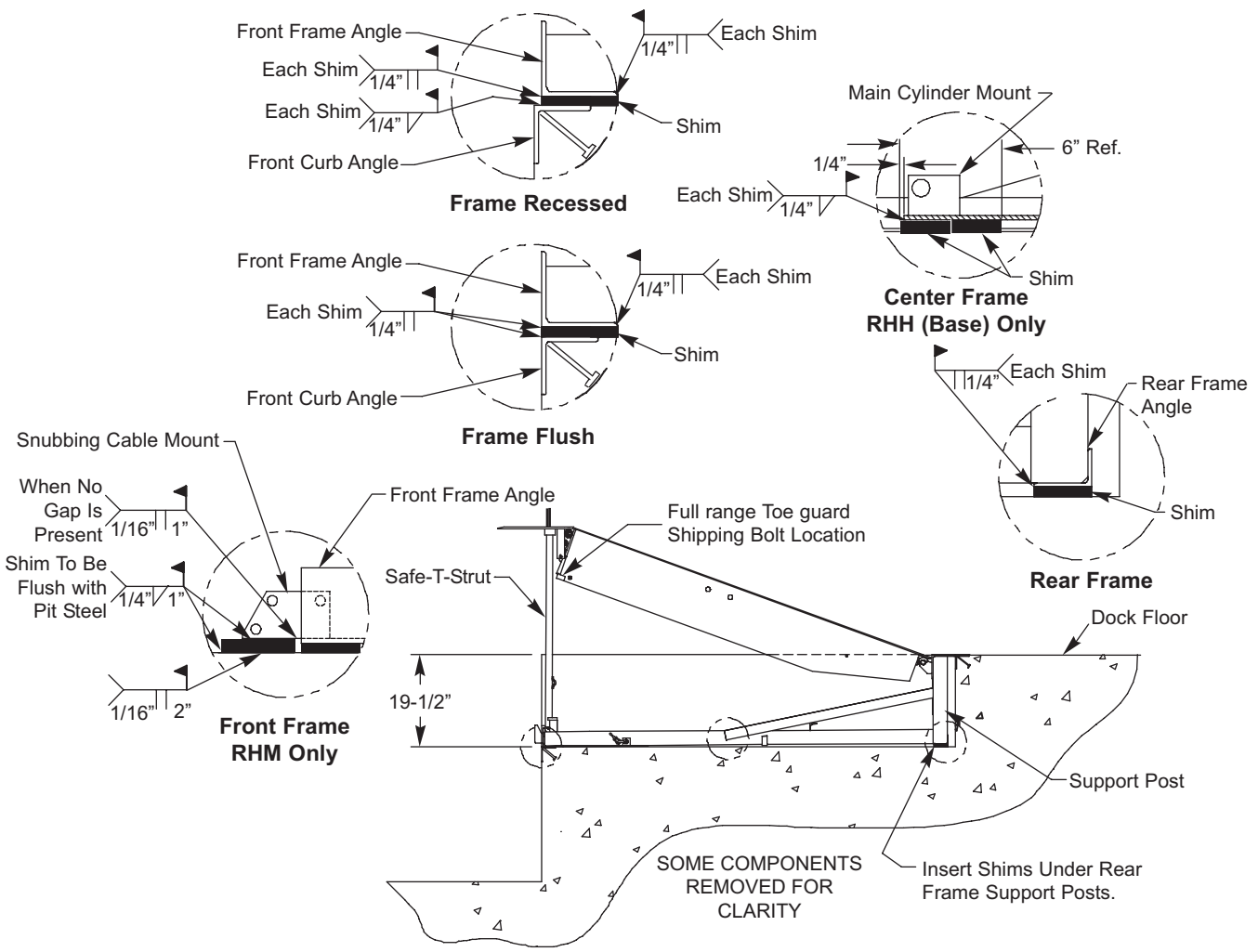


Figure - 10 Shimming

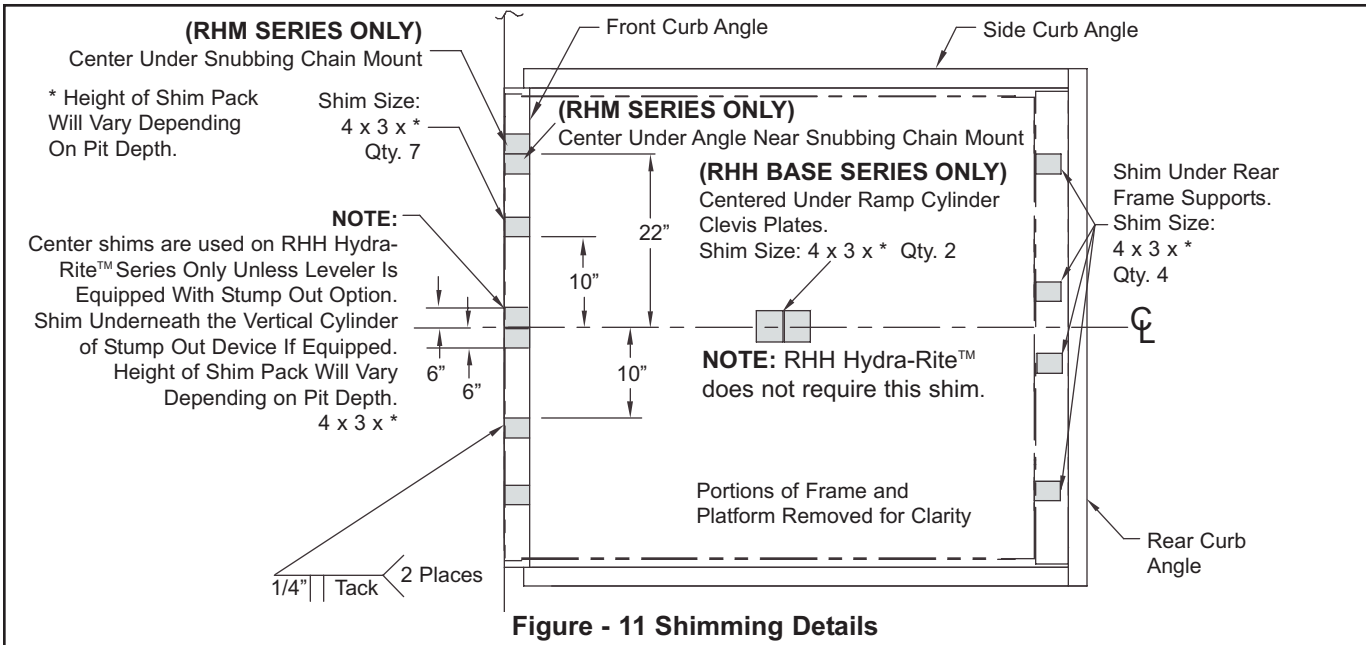


Figure - 11 Shimming Details

INSTALLATION INSTRUCTIONS CONT.

14. Check to insure leveler is centered and level in pit before final welding.
15. Stitch weld the rear frame angle (and rear spacers if needed) to the rear curb angle. See Figure 12.
16. Weld front frame angle to the front curb angle.
17. Clean and paint all weld and grind marks.
18. **RHA Series Only:** Insert platform attachment rods on each side of the Air Tower. Manually raise left side of the Air Tower and secure to J-hooks on platform stringer.
19. **RHA Series Only:** Raise right side of the Air Tower and ensure platform attachment rod is on top of the stringer anchor loops. Secure a wire tie around each anchor loop and the platform attachment rod.
20. **RHA Series Only:** Plug blower motor into pit outlet.
21. Remove front shipping bolts from each Toe guard if leveler is equipped with toe guards. Verify toe guards move up and down freely.
22. Complete Bumper and Lip Support installation procedures found in this manual.
23. Verify Lip Stop Bolt adjustment.
24. **Mechanical leveler:** Mount inside operation sign and Operate leveler several times and verify proper operation. Adjustments may be needed to insure proper operation. Adjust as needed. Reference Owners Manual for proper operation and adjustments.

    Powered leveler: Complete electrical installation. Operate leveler several times to verify proper operation. Adjustments may be needed to insure proper operation. Adjust as needed. Reference Owners Manual for proper operation and adjustments.

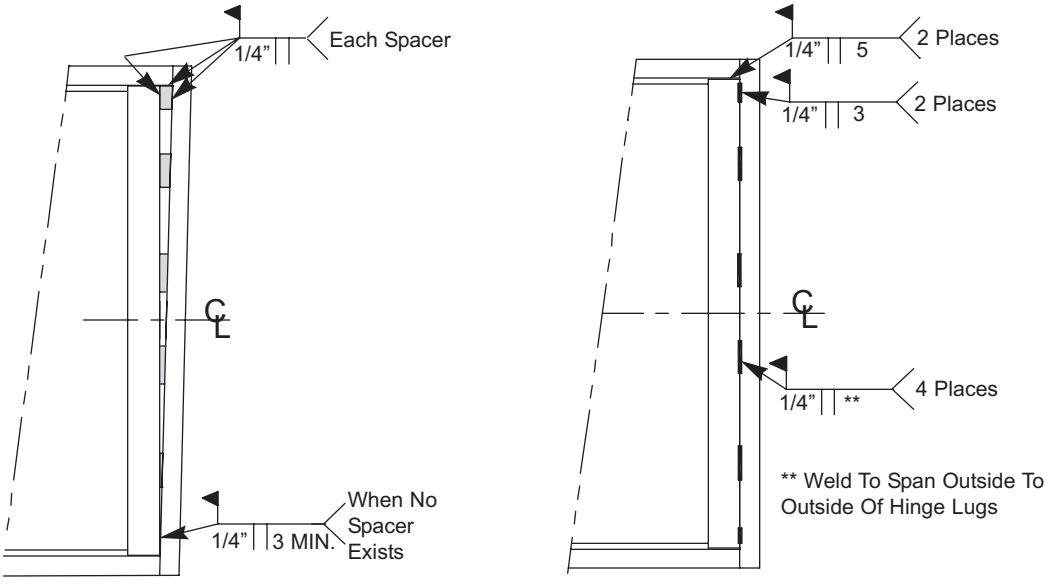


Figure - 12 Rear Weld Location

BUMPER INSTALLATION

LAMINATED BUMPER INSTALLATION (Standard)

1. Tack weld each bumper in position as shown in figure 14.
2. Each bumper should be secured with two 3/4 inch concrete anchor bolts that are a minimum of 5-1/4 inch long. (Ref. Rawl anchor 6954 - supplied by others.)
3. Using bumper mounting holes as a guide drill two 3/4 inch holes per bumper that are approximately 1/2 inch deeper than the bolt length that will be used.
4. Secure other end of each bumper with concrete anchor bolts. Torque each bolt to 120 ft-lbs.
5. Weld the bumper side plate to the front vertical pit angle.
6. Clean and paint all weld and grind marks.
7. Apply logo and serial number decal to front curb angle. See Figure 14.

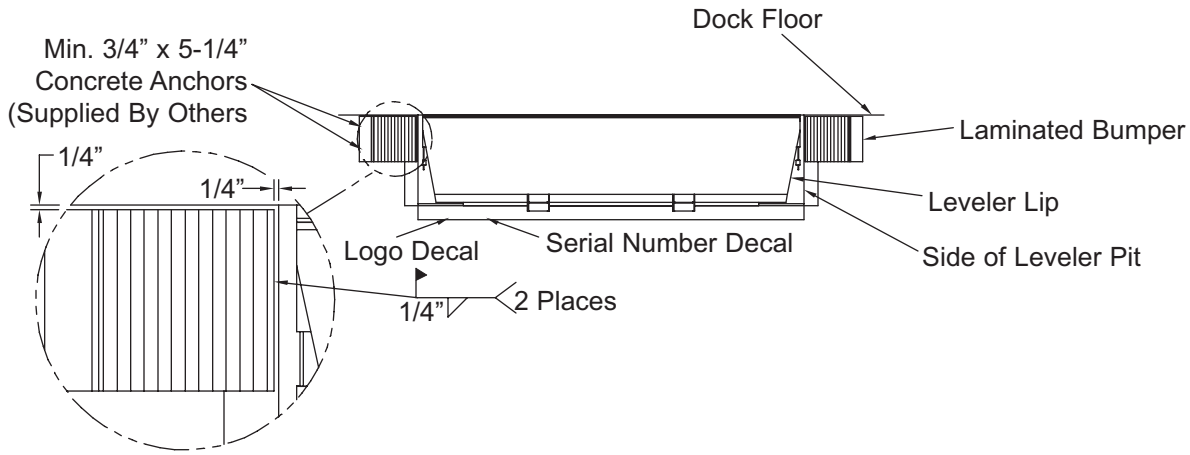


Figure - 14 Laminated Bumper Installation

LIP SUPPORT INSTALLATION

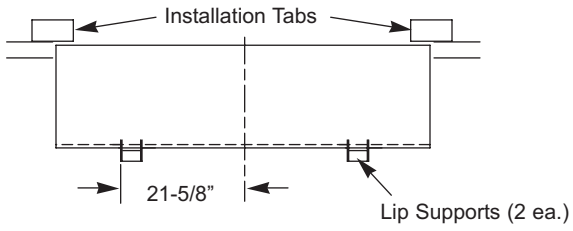


Figure - 15 Installation Tabs

NOTE: Powered Dock Levelers
It is suggested to add two installation tabs just behind the lip to support the leveler platform even and flush with the dock floor before lip support installation. See Figure 15.

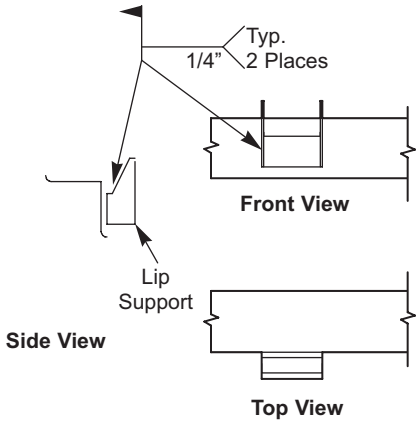
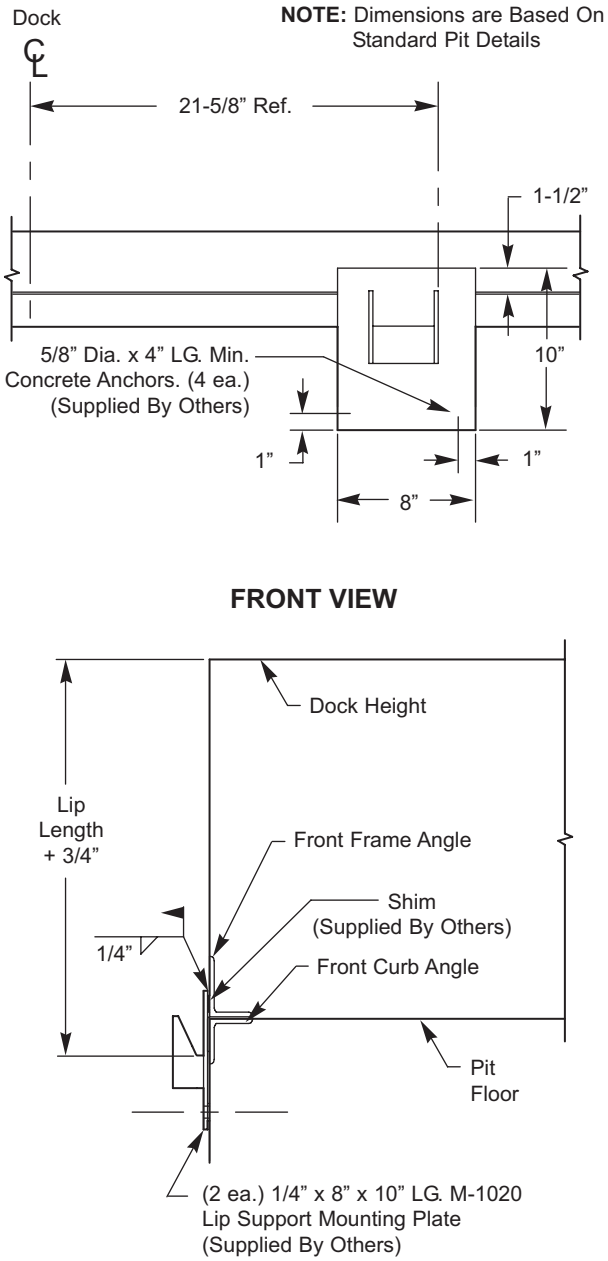


Figure - 16 Lip Support

1. Place marks 21-3/4" out from pit centerline on front pit angle. See Figure 15.
2. On levelers with lips greater than 20" install lip keeper mounting plate. See Figure 17.
3. Locate the centerline of the lip support assemblies.
4. Remove maintenance support and lower the leveler until it is flush with dock level and lip is fully pendant.
5. Position lip support assemblies along the edge of the lip and mark the bottom of lip support on front pit angle.
6. Fully weld along the lip support and front pit angle. See Figure 16.
7. Repeat the procedure for the second lip support assembly.
8. Clean and paint all weld and grind marks.



NOTE: Tack weld lip supports into place and check to insure leveler platform is even and flush with dock floor before final welding.

Figure - 17 Lip Mounting Plate

LIP STOP BOLT ADJUSTMENT

LIP STOP BOLT ADJUSTMENT

The lip stop bolt is factory adjusted prior to installation. Over time however, if the lip does NOT latch when the leveler is fully raised and remain latched as the platform lowers, the lip stop bolt may need to be readjusted. Use the following procedure to readjust the lip stop bolt:

1. Loosen the lock nuts on the lip stop bolt. See figure 18.

2. Turn the bolt clockwise until the gap between the back of the bolt head and the front lock nut is cut in half. See figure 18.

3. Tighten the lock nuts on the lip stop bolt.

4. Follow normal operating procedures to verify that the lip stop has been adjusted correctly and holds the lip out as the platform lowers.

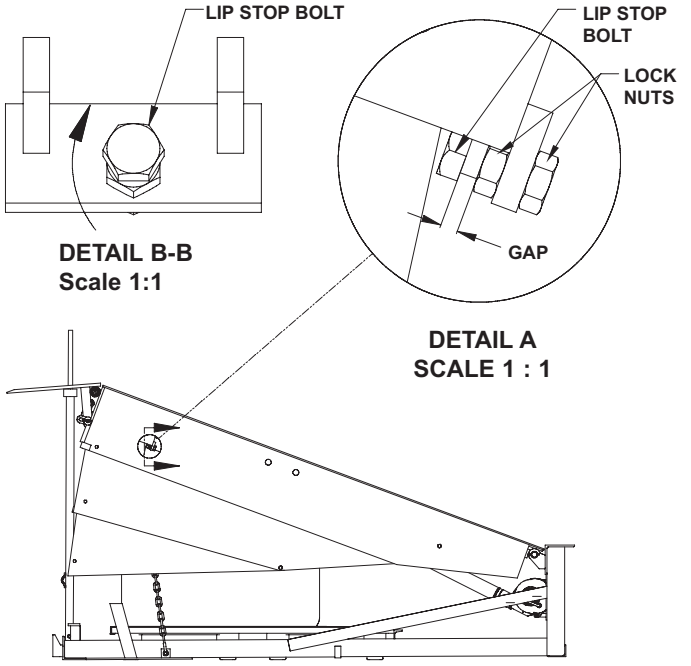


FIGURE 18 - LIP STOP BOLT ADJUSTMENT

TYPICAL COMPLETE INSTALLATION

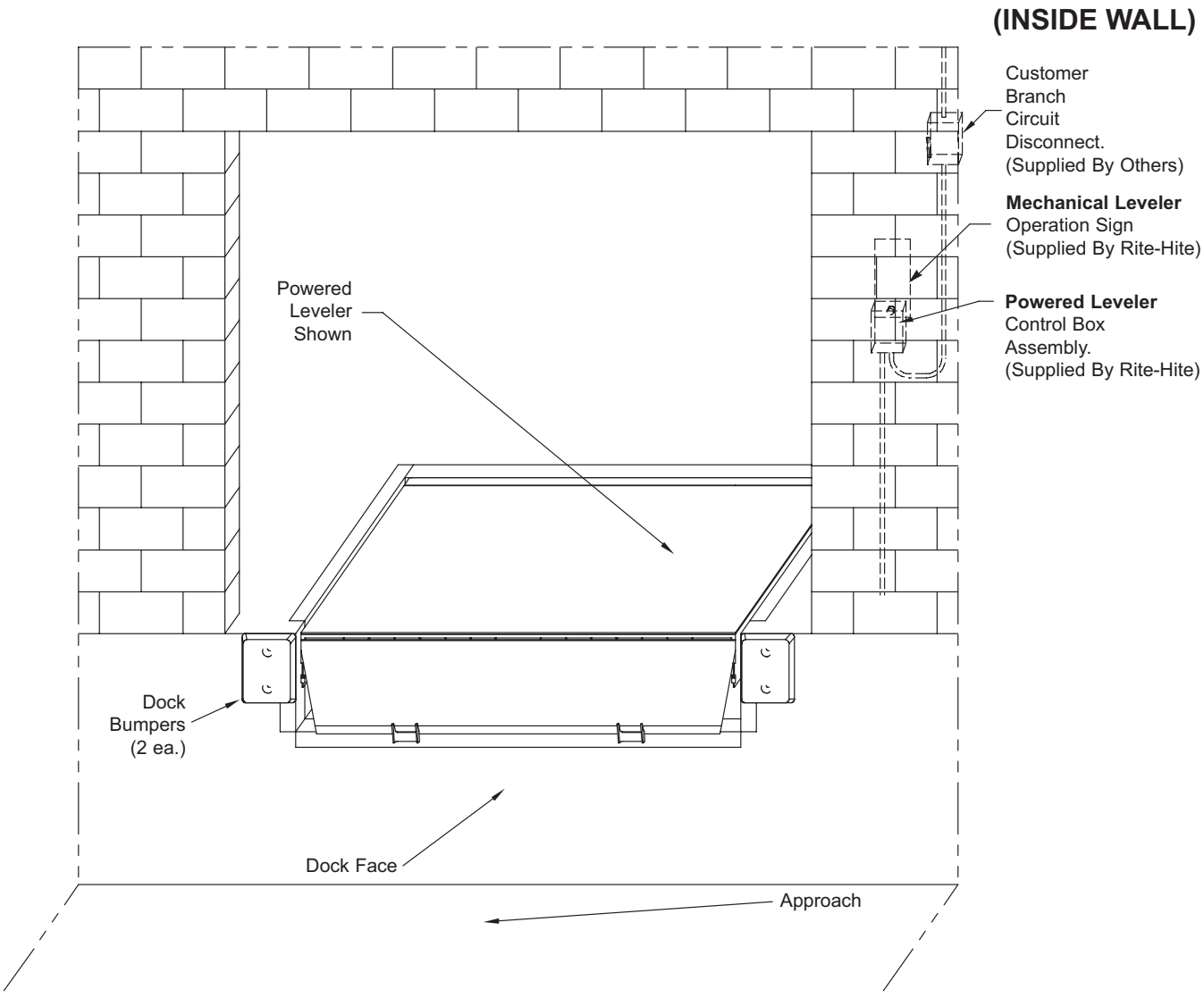


Figure - 19 Typical Installation

WARRANTY

RITE-HITE® STANDARD WARRANTY

RITE-HITE® warrants that its products will be free from defects in design, materials, and workmanship for a period of 365 days from the date of shipment. All claims for breach of this warranty must be made within 30 days after the defect is or can, with reasonable care, be detected and in no event no more than 30 days after the warranty has expired. In order to be entitled to the benefits of this warranty, the products must have been properly installed, maintained, and operated within their rated capacities and/or specified design parameters, and not otherwise abused. Periodic lubrication and adjustment is the sole responsibility of the owner. This warranty is RITE-HITE's® exclusive warranty. RITE-HITE® EXPRESSLY DISCLAIMS ALL IMPLIED WARRANTIES, INCLUDING THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS. Non-standard warranties, if any, must be specified by RITE-HITE® in writing.

In the event of any defects covered by this warranty, RITE-HITE® will remedy such defects by repairing or replacing any defective equipment or parts, bearing all the costs for parts, labor, and transportation. This shall be the exclusive remedy for all claims whether based on contract, negligence, or strict liability.

LIMITATION OF LIABILITY

RITE-HITE® SHALL NOT IN ANY EVENT BE LIABLE FOR ANY LOSS OF USE OF ANY EQUIPMENT OR INCIDENTAL OR CONSEQUENTIAL DAMAGES OF ANY KIND, WHETHER FOR BREACH OF WARRANTY, NEGLIGENCE, OR STRICT LIABILITY.

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